

# Reverb User Manual

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## AudioForge Reverb v1.0.0

Freeverb-inspired algorithmic reverb with room size and damping

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## Overview

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**Reverb** provides spacious reverberation with:

- **Room Size:** 0-100% (small room to cathedral)
  - **Damping:** 0-100% (bright to dark)
  - **Width:** 0-100% (mono to stereo)
  - **Wet/Dry Mix:** Blend reverb with dry signal
  - **Pre-Delay:** 0-100ms (early reflections)
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## Quick Start

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### Small Room Ambience

1. **Room Size:** 30%
2. **Damping:** 40%
3. **Mix:** 15%
4. Result: Tight, natural room sound

### Large Hall

1. **Room Size:** 80%

2. **Damping:** 60%
3. **Width:** 100%
4. **Mix:** 30%
5. Result: Spacious hall reverb

## Plate Reverb (Short + Bright)

1. **Room Size:** 40%
2. **Damping:** 20%
3. **Mix:** 25%
4. Result: Classic plate-like reverb

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## Parameters

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### Room Size

**Range:** 0-100%

**Description:** Simulated room dimensions

- **0-30%:** Small room, tight ambience
- **40-60%:** Medium room/studio
- **70-100%:** Large hall, cathedral

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### Damping

**Range:** 0-100%

**Description:** High-frequency absorption

- **0-30%:** Bright, reflective (tile, glass)
- **40-60%:** Neutral, balanced
- **70-100%:** Dark, absorptive (carpet, curtains)

## Width

**Range:** 0-100%

**Description:** Stereo width of reverb

- **0%:** Mono reverb (center)
  - **50%:** Moderate width
  - **100%:** Full stereo width
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## Mix

**Range:** 0-100%

**Description:** Wet/dry balance

- **10-20%:** Subtle ambience (vocals, leads)
  - **30-40%:** Moderate reverb (pads, guitars)
  - **50-100%:** Wet FX (send/return use)
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## Pre-Delay

**Range:** 0-100ms

**Description:** Delay before reverb starts

- **0ms:** Immediate reverb (thick, lush)
  - **20-40ms:** Slight separation (clarity)
  - **60-100ms:** Distinct echo + reverb (depth)
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## Use Cases

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### 1. Vocal Reverb (Natural)

**Settings:** Room 45%, Damping 50%, Width 70%, Mix 18%, Pre-Delay 30ms

**Result:** Clear, present vocals with space

### 2. Snare Reverb

**Settings:** Room 35%, Damping 40%, Width 80%, Mix 25%, Pre-Delay 0ms

**Result:** Punchy snare with natural ambience

### 3. Ambient Pad

**Settings:** Room 90%, Damping 70%, Width 100%, Mix 50%, Pre-Delay 10ms

**Result:** Lush, evolving ambient texture

### 4. Slapback + Reverb (Rockabilly)

**Settings:** Room 25%, Damping 30%, Mix 20%, Pre-Delay 80ms

**Result:** Short slap + subtle reverb tail

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## Tips & Techniques

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### 1. Pre-Delay for Clarity

Use 20-40ms pre-delay to separate direct sound from reverb (keeps vocals upfront).

### 2. Damping for Tone

- **Bright reverb** (low damping): Adds air and sparkle
- **Dark reverb** (high damping): Warm, vintage tone

### 3. Use as Send Effect

Set Mix to 100%, use on send/return for parallel reverb (better control).

### 4. EQ Before Reverb

Insert **SimpleEQ** before Reverb:

- HPF at 200 Hz (remove low-end mud in reverb)
- Slight boost at 5-8 kHz (airy reverb)

### 5. Automate Room Size

Automate Room Size from 30% → 90% during breakdown sections for dramatic builds.

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## Technical Specifications

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### Algorithm

- **Type:** Freeverb-inspired (Schroeder reverberator)
- **Comb Filters:** 8 parallel combs
- **Allpass Filters:** 4 in series
- **Stereo:** True stereo processing

### Performance

- **CPU:** 2-4% (moderate load)
  - **Latency:** ~5-10ms (algorithmic processing)
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# Troubleshooting

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## Reverb sounds muddy

**Fix:**

1. Increase Damping to 60-80%
2. Use HPF before Reverb (200-300 Hz)
3. Reduce Room Size slightly

## Too metallic/ringy

**Cause:** Excessive resonance in algorithm

**Fix:**

1. Reduce Room Size
2. Increase Damping
3. Lower Mix percentage

## Reverb kills punch

**Cause:** Too much wet signal, or no pre-delay

**Fix:**

1. Reduce Mix to 15-25%
2. Add Pre-Delay (20-30ms)
3. Use as send effect instead of insert

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### AudioForge Reverb v1.0.0

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